The *Robótica y Percepción Computacional* Final Project

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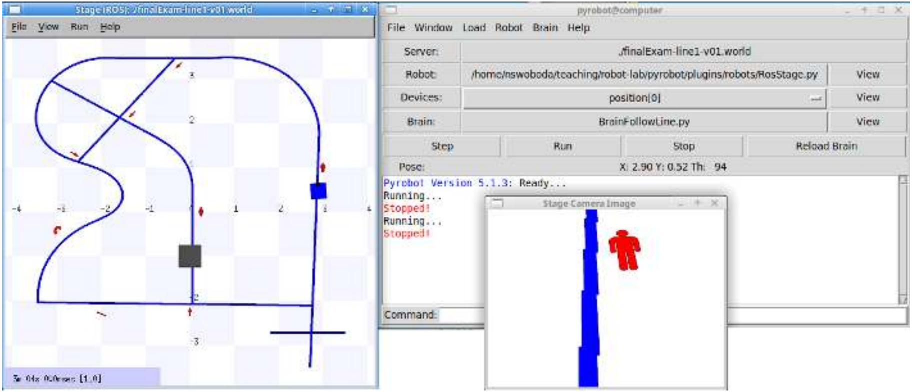
Project Overview

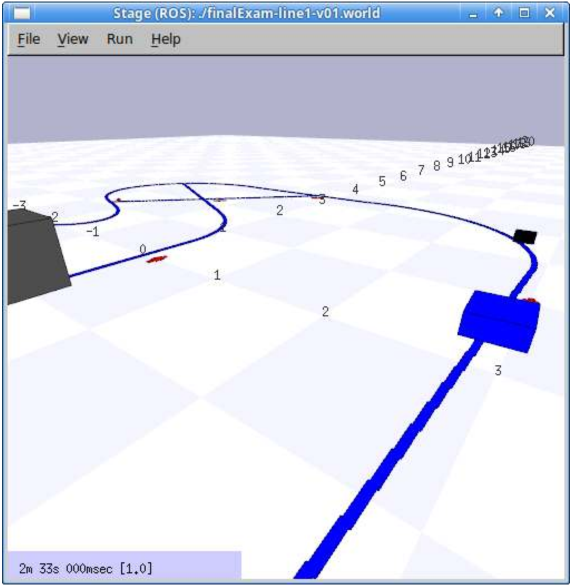
This course will focus on acquiring the theoretical and practical knowledge required to build the core modules of a *line following robot*.

To do so, you will be provided with both a virtual and a real environment for working with the robots.

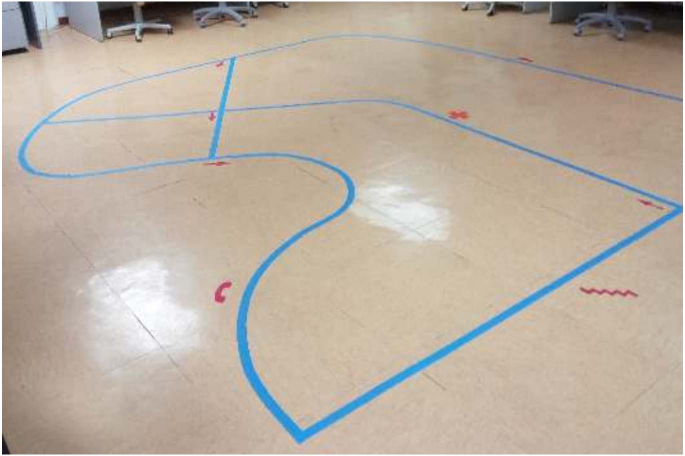
Project Example (virtual world)



Project Example (virtual world)



Project Example (real world)



(Show video)

The Final Project

The core modules that you will implement:

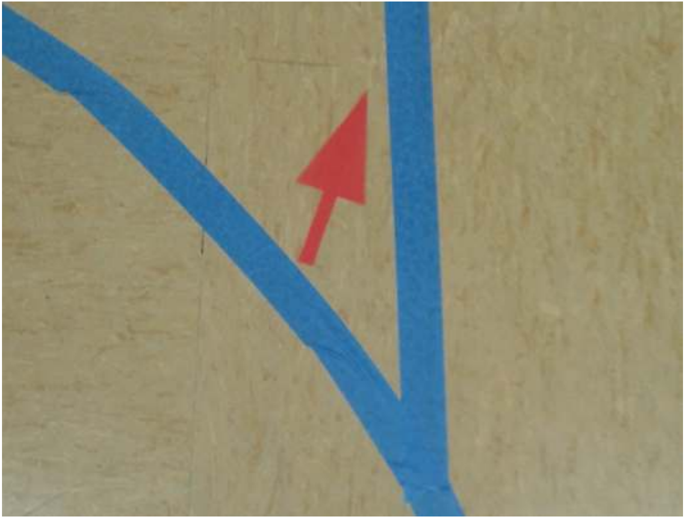
- Navigation
- Computer Vision
  - Image Segmentation
  - Symbol Recognition

In order to accomplish this, we will give you certain information, and we will make a number of simplifying assumptions.

The Final Project - Given information (1)

The lines, arrows, symbols and floor will always be certain known colors.

The arrows will always have the same shape



The Final Project - Given information (3)

The set of symbols will be known and they will always have the same shapes

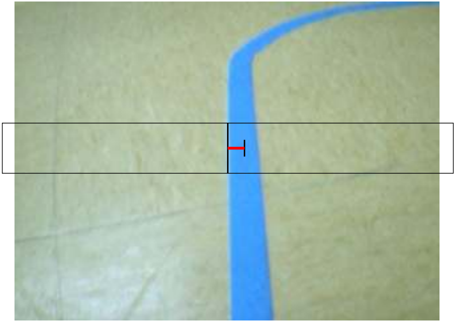
The Final Project - Simplifying Assumptions

Initially, we will make a number of assumptions to simplify the project:

- The robot initializes with the line in its field of vision
- The robot does not need to communicate with anyone/anything
- The robot's environment is all in the same plane
- No need to recharge
- ...

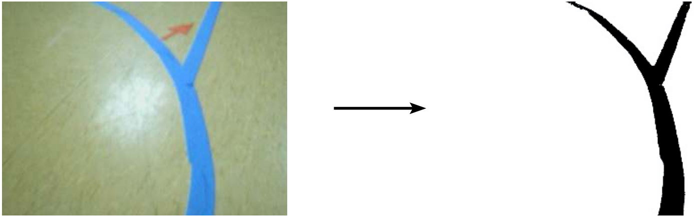
The Final Project - Navigation

You will implement a simple "reactive" control system to follow the line and avoid obstacles.



The Final Project - Segmentation

Using color models you will segment the video and identify intersections.

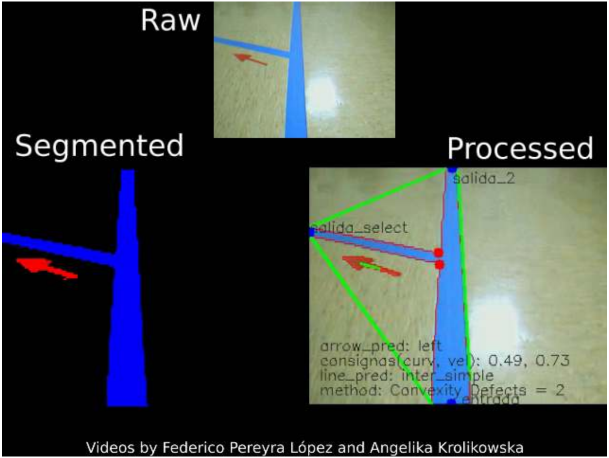


Using a segmented image of an arrow you will calculate its orientation.

The Final Project - Symbol Recognition

Using a segmented image of the symbols you will identify them.

The Final Project - Vision Example



(Show video)

The Final Project - Minimum Requirements

The implemented *real world system* must at least be able to:

- Follow a simple blue line
- Identify "Y-type" intersections in the line and follow arrows to chose the appropriate branch
- Recognize and announce the location of at least 2 symbols.

Only students who demo their final projects with a real robot will have the chance of receiving the maximum grade (10/10).

Groups who demo their final projects with a simulated robot will have slightly different minimum requirements and will receive at most a 7/10.

After implementing the "basic system" each group can choose how to extend their system to include more functionality. For example:

- more complicated types of line intersections (very sharp turns, . . .)
- capability to search for the line when not found at the start of a run or when the line is lost
- avoiding obstacles
- problems with line discontinuity (breaks) or noise (dirt or reflections from the lights)
- use a homemade robot
- . . .

The Final Exam

On the day of the final exam, each group will need to demonstrate their system:

- The robot will need to follow two different routes:
  - The first route will be known in advance
  - The second route will be announced on the day of the exam
- During 15 minutes you can rerun the routes as many times as you like

The Course's Lab Space